Majorana Modes in the Machine

Week 8: NISQ Reality Check – Noise Model and Failure Modes

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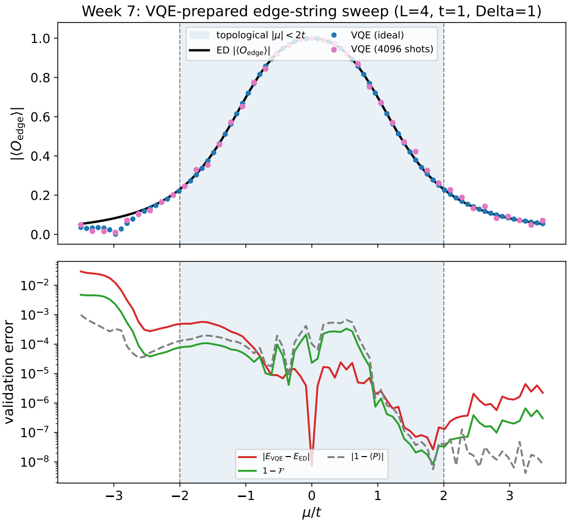
What Week 7 Gives Us



Noiseless baseline

- Prepared by parity-constrained VQE
- Edge string measured ideally and with shots
- ED validates energy, parity, fidelity, string value

What Noise Can Break

State preparation

- VQE ansatz gates no longer implement exactly $U(\theta)$.
- Entangling-gate errors reduce edge coherence.
- Relaxation can bias the parity sector.

Measurement

- Readout flips corrupt the bitstring parity product.
- Shot noise broadens the string estimator.
- A small true signal becomes hard to distinguish from zero.

Main risk

Noise can flatten the edge-string contrast between $|\mu| < 2t$ and $|\mu| > 2t$, even when the noiseless circuit is correct.

Noise Channels to Add

Channel	Physical meaning	Expected effect
Depolarizing error	Random Pauli error after gates	Shrinks correlations and string contrast
Readout error	Classical bit flip during measurement	Biases ± 1 parity-product estimator
Amplitude damping	T_1 relaxation toward $\|0\rangle$	Can change particle number and parity balance
Phase damping	T_2 dephasing	Destroys coherent edge-to-edge correlations

Two Experiments to Keep Separate

Frozen-parameter noise test

- Use $\theta^*(\mu)$ from noiseless Week 7.
- Add noise only during circuit execution.
- Measures readout and circuit degradation at fixed preparation.

Noisy optimization test

- Optimize the VQE cost under noisy estimates.
- Tests whether the optimizer can still find good parameters.
- Harder and more expensive; should come second.

Engineering rule

First isolate measurement/circuit degradation; only then study optimizer degradation.

Circuit-Level Noise Protocol: Formalism

Objective: Formalize the mapping from the noiseless VQE sweep to the noisy density matrix.

For the Week 7 optimized parameters $\theta^*(\mu)$, the ideal pure state is $\rho_{\text{id}} = |\psi(\theta^*)\rangle\langle\psi(\theta^*)|$. We define the physical execution via completely positive trace-preserving (CPTP) maps:

$$\rho_{\text{noisy}} = \mathcal{E}_{\text{readout}} \circ \mathcal{E}_{\text{meas-basis}} \circ \mathcal{E}_{\text{gate}}(\rho_{\text{id}}) \tag{1}$$

- **Gate Noise ($\mathcal{E}_{\textbf{gate}}$):** Local two-qubit depolarizing channels after entangling gates, $\mathcal{D}_p(\rho) = (1-p)\rho + \frac{p}{15}\sum_{i=1}^{15} P_i \rho P_i$, over the non-identity two-qubit Paulis.
- **Basis Rotation ($\mathcal{E}_{\textbf{meas-basis}}$):** Ideal unitary mapping for $X_0 Z_1 Z_2 X_3$.
- **Readout Assignment ($\mathcal{E}_{\textbf{readout}}$):** Asymmetric bit-flip channel $p(0|1)$ and $p(1|0)$ applied to the diagonal elements of the density matrix.

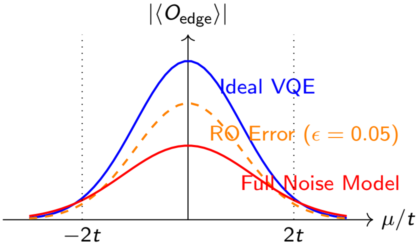
Frozen-Parameter Test: Readout vs. Depolarizing Channels

Protocol: Evaluate $\mathrm{Tr}(\hat{O}_{\mathrm{edge}}\rho_{\mathrm{noisy}})$ using frozen $\theta^*(\mu)$ from the ED/ideal VQE baseline.

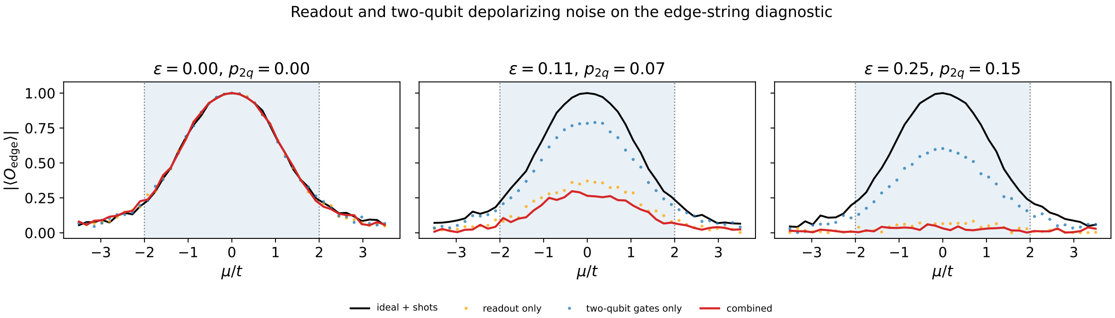
Expected Behavior:

- The ideal ED baseline peaks at $\langle O_{\mathrm{edge}}\rangle = 1.0$ at $\mu = 0$.
- *Readout error alone* scales the curve by $(1-2\epsilon)^4$ for independent symmetric readout flips on the four measured bits.
- *Gate noise* lowers purity, $\mathrm{Tr}(\rho^2) < 1$, and suppresses coherent edge correlations, especially near $\mu_c = \pm 2t$.

Schematic expectation before simulator data



Visualising the Noise



Same Kitaev/Jordan-Wigner Hamiltonian and edge-string operator as Block 3; only the execution noise is varied.

Phase-Boundary Failure Criterion

We must define mathematically when the topological transition at $|\mu| = 2t$ becomes indistinguishable from the trivial vacuum due to the combined noise channels.

Failure Criterion: The diagnostic fails when the measured contrast between the topological window and the trivial region cannot be resolved from the statistical and hardware noise floor with a 3σ confidence interval.

Specifically, failure occurs if the measurable contrast satisfies:

$$\max_{|\mu|<2t} |\langle \hat{O}_{\mathsf{edge}}(\mu)\rangle_{\mathsf{meas}}| - \max_{|\mu|>2t} |\langle \hat{O}_{\mathsf{edge}}(\mu)\rangle_{\mathsf{meas}}| \leq \frac{3}{\sqrt{N_{\mathsf{shots}}}} + \delta_{\mathsf{SPAM}} \tag{2}$$

where δ_{SPAM} is the systematic bias inferred from the parity drift observable, $|1 - \langle \hat{P} \rangle_{\mathsf{noisy}}|$.

Next Week

Next milestone

First produce the noisy frozen-parameter sweep against the Week 7 baseline; then run noisy VQE optimization and compare raw versus mitigated readout.